Clinical NLP for RCT Abstract Sentence Classification: A Comparison of Bidirectional LSTM and DistilBERT

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Project Information

GitHub repository: `https://github.com/GG1627/bme6938-project-3`

Team members and contributions:

Gael Garcia – model architecture, LSTM and DistilBERT training pipeline, GitHub setup

Jada Brown – data preprocessing, EDA notebook, class imbalance analysis, visualizations

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Abstract

The rapid growth of biomedical literature has made manual review of randomized controlled trial (RCT) abstracts increasingly impractical for clinicians and researchers. Automatically classifying sentences in RCT abstracts into their rhetorical roles (Background, Objective, Methods, Results, and Conclusions) can accelerate evidence synthesis, support structured summarization, and enable downstream clinical decision support. In this work, we train and compare two neural models on the PubMed RCT 20k dataset [1], a benchmark corpus of 235,892 labeled sentences from clinical trial abstracts. The models we used are a bidirectional LSTM baseline and a fine-tuned DistilBERT transformer [3]. The LSTM model achieves 84% test accuracy with a macro F1 of 0.78 and 6.2M trainable parameters, while DistilBERT achieves 86% test accuracy with a macro F1 of 0.80 and 67M parameters. Both models struggle most on the minority class (Objective, 7.8% of data), with the background-objective boundary representing the most common failure mode across both architectures. Our results demonstrate that a lightweight recurrent baseline can approach transformer-level performance on this task, and that class imbalance remains a key challenge for clinical NLP sentence classification regardless of model complexity.

Index Terms

clinical natural language processing, sentence classification, randomized controlled trial, bidirectional LSTM, DistilBERT, PubMed RCT, evidence synthesis, biomedical text mining

I. Introduction

The volume of published biomedical literature has grown at an unprecedented rate over the past several decades, with PubMed indexing more than 35 million citations as of 2024. Randomized controlled trials (RCTs) represent the highest level of clinical evidence, and their abstracts serve as the primary entry point for clinicians, researchers, and systematic reviewers seeking to evaluate treatment efficacy and safety. However, the sheer volume of RCT literature makes manual screening and synthesis increasingly infeasible. A clinician or researcher attempting to review hundreds of abstracts must repeatedly parse unstructured text to locate specific types of information, such as the study objective, the methods used, or the reported outcomes. This process is time consuming, prone to error, and difficult to scale.

Structured abstracts, which organize content under labeled headings such as Background, Objective, Methods, Results, and Conclusions, have been shown to improve readability and information retrieval in biomedical literature

[9]. However, a large proportion of published RCT abstracts remain unstructured or inconsistently labeled across journals and publishing venues. Automatically classifying each sentence in an RCT abstract into its rhetorical role would allow for unstructured abstracts to be turned into structured representations that supports downstream applications such as evidence synthesis, clinical guideline generation, and literature-based question answering.

Natural language processing (NLP) has emerged as a critical tool for extracting structured information from biomedical text [10]. Early approaches to abstract sentence classification relied on handcrafted features and shallow machine learning models such as naive Bayes and conditional random fields. The introduction of deep learning methods, particularly recurrent neural networks and later transformer-based architectures, has substantially improved performance on this task. The PubMed RCT dataset introduced by Dernoncourt and Lee [1] established a large-scale benchmark for sequential sentence classification in medical abstracts and enabled systematic comparison of neural approaches. Subsequent work by Jin and Szolovits [7] demonstrated that hierarchical neural networks which model inter-sentence context can further improve classification performance, while more recent work has explored the use of pretrained language models for this task [16].

Despite this progress, several challenges remain. Class imbalance across rhetorical roles, the boundary ambiguity between semantically similar classes such as Background and Objective, and the computational cost of large transformer models all present practical barriers to clinical deployment. Lightweight models that can approach transformer-level performance with a fraction of the parameters would be preferable in resource-constrained clinical settings.

This project addresses these challenges by training and comparing two neural models on the PubMed RCT 20k dataset [1], a bidirectional LSTM baseline and a fine-tuned DistilBERT model [3]. Our goals are to (1) establish a strong recurrent baseline for this task, (2) evaluate the performance gain achieved by fine-tuning a pretrained transformer, (3) analyze the effect of class imbalance on per-class performance, and (4) identify the primary failure modes shared across both architectures. The intended beneficiaries of this work include systematic reviewers, clinical informaticists, and evidence-based medicine platforms seeking to automate or accelerate the processing of RCT literature.

II. Literature Review

The task of classifying sentences in biomedical abstracts into rhetorical roles has a well-established history in clinical NLP research. Early approaches relied on rule-based and shallow machine learning methods, including naive Bayes classifiers, hidden Markov models, and conditional random fields applied to handcrafted linguistic features such as part-of-speech tags, word frequency distributions, and sentence position within the abstract. While these methods established the feasibility of the task, their reliance on manual feature engineering limited both their scalability and their ability to generalize across different journal styles and clinical domains [10].

The release of the PubMed 200k RCT dataset by Dernoncourt and Lee [1] marked a turning point for this research area. By providing a large-scale labeled corpus of approximately 200,000 RCT abstracts with sentence-level rhetorical role annotations, the dataset enabled systematic benchmarking of deep learning approaches for sequential sentence classification in biomedical text. The authors also introduced a recurrent neural network baseline using a bidirectional LSTM with a conditional random field (CRF) output layer, demonstrating that recurrent architectures could capture sequential dependencies between sentences within an abstract and substantially outperform shallow baselines on this task [1].

Building on this foundation, Jin and Szolovits [7] proposed a hierarchical sequential labeling network that explicitly models inter-sentence context. Their model encodes each sentence independently using a convolutional or recurrent layer, then applies a second bidirectional LSTM over the sequence of sentence representations to capture discourse-level structure within the abstract. This hierarchical approach outperformed prior state-of-the-art results by 2 to 3 percent on both the PubMed 200k and 20k RCT benchmarks, demonstrating that contextual information across sentences is highly informative for this classification task [7]. Our work differs from Jin and Szolovits in that we classify each sentence independently rather than jointly, which reduces model complexity and better reflects a practical deployment scenario where abstracts may be processed incrementally.

The introduction of the Transformer architecture by Vaswani et al. [4] fundamentally changed the landscape of NLP by replacing recurrence with self-attention mechanisms, enabling models to capture long-range dependencies

in text more efficiently and with greater parallelism. BERT [2], introduced by Devlin et al., extended this framework through large-scale unsupervised pretraining on general-domain text, producing contextual word representations that can be fine-tuned for a wide range of downstream tasks with minimal task-specific architectural modifications. BERT and its variants have since achieved state-of-the-art performance across numerous biomedical NLP benchmarks.

Domain-specific pretraining has proven particularly valuable in biomedical NLP. Lee et al. introduced BioBERT [6], a version of BERT pretrained on large biomedical corpora including PubMed abstracts and PubMed Central full-text articles. BioBERT substantially outperforms general-domain BERT on biomedical named entity recognition, relation extraction, and question answering tasks, demonstrating that the vocabulary and semantic patterns of biomedical text are sufficiently distinct from general English to warrant domain-specific pretraining. Cohan et al. [16] further demonstrated that pretrained language models fine-tuned on the PubMed RCT task can achieve significantly higher F1 scores than recurrent baselines, establishing transformers as the current state of the art for abstract sentence classification.

For this project, we chose DistilBERT [3] rather than a domain-specific model such as BioBERT as our transformer baseline. DistilBERT retains 97 percent of BERT's language understanding capability while reducing model size by 40 percent and improving inference speed by 60 percent through knowledge distillation [3]. Given that the PubMed RCT dataset contains clean, structured scientific English rather than dense clinical jargon, we hypothesized that the domain pretraining advantage of BioBERT would be minimal for this specific task, while the efficiency advantage of DistilBERT would be practically meaningful. This choice is also consistent with the goal of identifying lightweight models suitable for resource-constrained clinical deployment settings.

The ethical dimensions of deploying NLP models in clinical contexts have received increasing attention in recent literature. Bear Don't Walk et al. [11] conducted a scoping review of ethics considerations specific to clinical NLP, identifying bias in training data, lack of model transparency, and risks of perpetuating health disparities as primary concerns. Obermeyer et al. [12] demonstrated empirically that biases embedded in training data can propagate through AI systems in ways that disproportionately affect underserved patient populations, underscoring the importance of careful dataset curation and evaluation in any clinical NLP application. These considerations directly inform our discussion of limitations and ethical considerations in Section V.

While prior work has focused primarily on maximizing classification performance through architectural innovations and domain-specific pretraining, this project contributes a direct empirical comparison between a lightweight recurrent baseline and a general-purpose pretrained transformer on the PubMed RCT 20k task. By analyzing per-class performance, failure modes, and the efficiency tradeoff between model size and accuracy, we aim to provide practical guidance for researchers and developers considering NLP-based abstract classification tools in real-world clinical and biomedical settings.

III. Methods & Data

A. Dataset

This project uses the PubMed RCT 20k dataset [1], a publicly available benchmark corpus hosted on Hugging Face and originally introduced by Dernoncourt and Lee for sequential sentence classification in medical abstracts. The dataset contains 235,892 sentences drawn from randomized controlled trial abstracts indexed on PubMed, with each sentence labeled according to its rhetorical role in the abstract. The five label classes are Background, Objective, Methods, Results, and Conclusions. Pre-built train, validation, and test splits are provided, containing 176,642, 29,672, and 29,578 sentences respectively. Each example includes the sentence text, its label, the abstract identifier, and a sentence position index within the abstract.

The dataset exhibits moderate class imbalance. Methods (33.6%) and Results (32.8%) sentences together account for over 66% of the training set, while Objective sentences represent only 7.8% of training examples. This imbalance is consistent across all three splits, indicating that it reflects the natural distribution of rhetorical roles in RCT literature rather than a data collection artifact. All numbers in the dataset have been replaced with the token `@`, a preprocessing convention applied uniformly across the corpus that anonymizes specific numeric values while preserving the structural context in which they appear.

B. Exploratory Data Analysis

Prior to modeling, we conducted an exploratory data analysis to characterize the dataset and inform preprocessing decisions. Class distribution plots confirmed the imbalance described above and motivated the use of class-weighted loss during training. Sentence length analysis showed that the median word count across all classes ranges from 22 to 26 tokens, with a 95th percentile of 55 tokens and a maximum of 296 tokens. This distribution informed our choice of a maximum sequence length of 128 tokens for both models, which captures over 95% of sentences without wasting compute on excessive padding. Qualitative inspection of sample sentences revealed distinct linguistic patterns across classes: Background sentences tend to reference prior work and general clinical context, Objective sentences use purpose-driven language, Methods sentences are procedural and contain dense `@` tokens representing anonymized numeric values, Results sentences are statistically oriented, and Conclusions sentences frequently contain recommendation language. These patterns suggested that both models would have sufficient linguistic signal to learn meaningful class boundaries.

C. Preprocessing

Text preprocessing was kept minimal and intentional. All text was lowercased for the LSTM model to reduce vocabulary size, while the DistilBERT tokenizer handled its own normalization internally. The `@` token was retained in both pipelines, as it carries meaningful distributional signal, appearing most frequently in Methods and Results sentences. Integer labels were encoded using a fixed mapping: Background (0), Objective (1), Methods (2), Results (3), and Conclusions (4).

To address class imbalance, we computed class weights using scikit-learn's `compute_class_weight` function with the balanced setting [13], which scales each class weight inversely proportional to its frequency in the training set. The resulting weights were Background (1.92), Objective (2.55), Methods (0.60), Results (0.61), and Conclusions (1.30), and were passed directly to the cross-entropy loss function during training for both models.

D. LSTM Baseline

The recurrent baseline is a bidirectional LSTM classifier [5] implemented in PyTorch [14]. A vocabulary of 30,000 tokens was constructed from the training set using word frequency counts, with index 0 reserved for padding and index 1 for unknown tokens. Input sentences are tokenized by whitespace, truncated or padded to a maximum length of 128 tokens, and encoded as sequences of vocabulary indices.

The model architecture consists of a learnable embedding layer of dimension 128, followed by a two-layer bidirectional LSTM with hidden dimension 256 per direction. The final hidden states from the forward and backward passes are concatenated to produce a 512-dimensional sentence representation, which is passed through a dropout layer with probability 0.3 and a fully connected classification head mapping to 5 output logits. The total number of trainable parameters is 6.2 million. The model was trained using the Adam optimizer [4] with a learning rate of 1e-3, a batch size of 64, gradient clipping at a maximum norm of 1.0, and a ReduceLROnPlateau learning rate scheduler with patience 2 and reduction factor 0.5. Early stopping was applied with patience 5, saving the checkpoint with the lowest validation loss. Training terminated at epoch 10 with the best checkpoint saved at epoch 5.

E. DistilBERT Model

The transformer model is DistilBERT [3], a distilled version of BERT [2] that retains 97% of BERT's language understanding capability at 40% of the model size. We fine-tuned the `distilbert-base-uncased` checkpoint available through the Hugging Face Transformers library [8] for sequence classification with 5 output classes. The DistilBERT tokenizer was used to encode all sentences with a maximum length of 128 tokens, using padding and truncation as needed. All 66,957,317 parameters of the model were fine-tuned end-to-end.

Training used the Adam optimizer with a learning rate of 2e-5, a batch size of 128, gradient clipping at a maximum norm of 1.0, and a ReduceLROnPlateau scheduler with patience 1 and reduction factor 0.5. The same class-weighted cross-entropy loss used for the LSTM was applied. Early stopping with patience 3 was used, and the best checkpoint by validation loss was saved. Training terminated at epoch 5 with the best checkpoint at epoch 2.

F. Evaluation Strategy

Both models were evaluated on the same held-out test set of 29,578 sentences. Reported metrics include overall accuracy, per-class precision, recall, and F1-score, macro-averaged F1, and weighted-averaged F1. Macro F1 is treated as the primary comparison metric, as it gives equal weight to all classes regardless of support and is therefore more sensitive to performance on minority classes such as Objective. All experiments were conducted on a single NVIDIA A100 GPU on the University of Florida HiPerGator computing cluster.

G. AI Assistance Disclosure

Claude (Anthropic) was used as a coding assistant during development of the training pipeline and notebooks. All code was reviewed, tested, and understood by the authors. AI assistance is documented per course guidelines.

IV. Results & Evaluation

A. LSTM Baseline Performance

The bidirectional LSTM baseline was trained for 10 epochs before early stopping triggered, with the best checkpoint saved at epoch 5. Table I presents the full classification report on the held-out test set of 29,578 sentences.

TABLE I
LSTM Test Set Performance

Class	**Precision**	**Recall**	**F1**	**Support**
Background	0.61	0.70	0.65	3,077
Objective	0.69	0.60	0.64	2,333
Methods	0.91	0.93	0.92	9,884
Results	0.92	0.88	0.90	9,713
Conclusions	0.78	0.80	0.79	4,571
Accuracy	0.84			29,578
Macro avg	0.78	0.78	0.78	29,578
Weighted avg	0.84	0.84	0.84	29,578

The LSTM achieves 84% overall accuracy and a macro F1 of 0.78. Performance is strongest on the Methods (F1 0.92) and Results (F1 0.90) classes, which together comprise over 66% of the test set and exhibit the most distinctive linguistic patterns. The Objective class achieves an F1 of 0.64, reflecting the difficulty of classifying the minority class with only 7.8% of training examples. Background achieves an F1 of 0.65, with the confusion matrix revealing that a substantial proportion of Background sentences are misclassified as Objective, consistent with the lexical similarity between these two intro-style sentence types.

B. DistilBERT Performance

The DistilBERT model was trained for 5 epochs before early stopping triggered, with the best checkpoint saved at epoch 2. Table II presents the full classification report on the same held-out test set.

TABLE II
DistilBERT Test Set Performance

Class	**Precision**	**Recall**	**F1**	**Support**
Background	0.65	0.73	0.69	3,077
Objective	0.68	0.64	0.66	2,333
Methods	0.93	0.93	0.93	9,884
Results	0.93	0.88	0.91	9,713
Conclusions	0.79	0.85	0.82	4,571
Accuracy	0.86			29,578
Macro avg	0.80	0.81	0.80	29,578
Weighted avg	0.86	0.86	0.86	29,578

DistilBERT achieves 86% overall accuracy and a macro F1 of 0.80, improving over the LSTM baseline by 2 percentage points in accuracy and 0.02 in macro F1. Gains are consistent across Background (F1 0.69 vs 0.65), Objective (F1 0.66 vs 0.64), Methods (F1 0.93 vs 0.92), Results (F1 0.91 vs 0.90), and Conclusions (F1 0.82 vs 0.79). The Objective class remains the weakest for both models, confirming that class imbalance is the dominant challenge for this task regardless of architecture.

C. Model Comparison

Table III provides a direct side-by-side comparison of both models across all evaluation metrics, including model size.

TABLE III
MODEL COMPARISON SUMMARY

Metric	**LSTM**	**DistilBERT**
Accuracy	0.84	0.86
Macro F1	0.78	0.80
Weighted F1	0.84	0.86
Background F1	0.65	0.69
Objective F1	0.64	0.66
Methods F1	0.92	0.93
Results F1	0.90	0.91
Conclusions F1	0.79	0.82
Parameters	6.2M	67M
Train time/epoch	$\sim$2 min	$\sim$12 min

DistilBERT outperforms the LSTM on every metric, but the margin is modest given the 10x difference in parameter count and 6x difference in training time per epoch. This suggests that the task is learnable with relatively simple sequence representations, and that the contextual embeddings produced by DistilBERT provide incremental rather than transformative gains on clean, structured biomedical text.

D. Confusion Matrix Analysis

Figure 1 shows the confusion matrix for the LSTM model and Figure 2 shows the confusion matrix for DistilBERT on the test set. In both cases, the diagonal is strongly dominant, confirming that both models classify the majority of sentences correctly. The most prominent off-diagonal entries in both matrices are Background sentences misclassified as Objective and Objective sentences misclassified as Background. This is consistent with the qualitative observation from our EDA that both classes use introductory language and general clinical framing, making them difficult to distinguish without broader abstract-level context.

E. Training Curves

Figure 3 shows the training and validation loss and accuracy curves for both models. The LSTM exhibits clear overfitting after epoch 5, with training accuracy continuing to rise while validation loss diverges. DistilBERT converges more quickly, with the best validation checkpoint reached at epoch 2 and overfitting evident from epoch 3 onward. Both curves confirm that early stopping and best-checkpoint saving were necessary to prevent performance degradation on the validation and test sets.

F. Per-Class F1 Comparison

Figure 4 shows a grouped bar chart comparing per-class F1 scores between the LSTM and DistilBERT. DistilBERT achieves higher F1 on all five classes, with the largest absolute gains on Background (0.04) and Conclusions (0.03). The smallest gain is on Methods (0.01), where both models already perform near ceiling given the strong and distinctive linguistic patterns in that class.

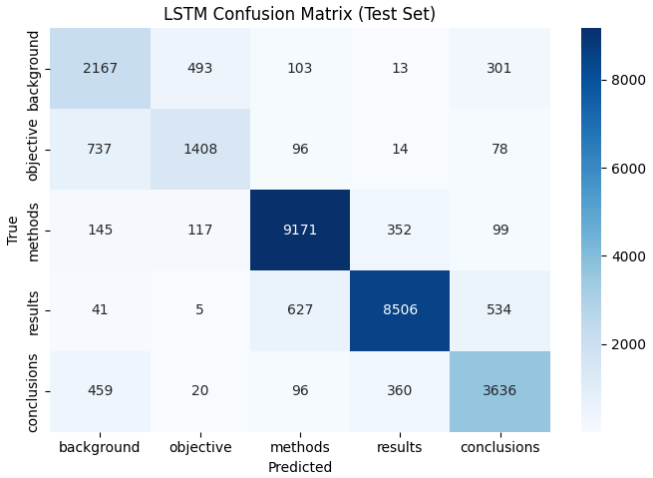

Fig. 1. Confusion matrix for the LSTM baseline on the test set. The most common errors occur at the Background-Objective boundary, where both classes share introductory sentence structure.

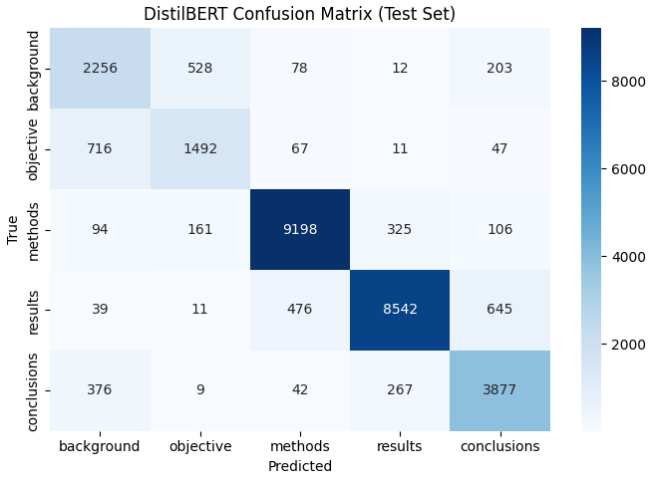

Fig. 2. Confusion matrix for DistilBERT on the test set. DistilBERT reduces Background-Objective confusion relative to the LSTM, but the boundary remains the primary failure mode for both models.

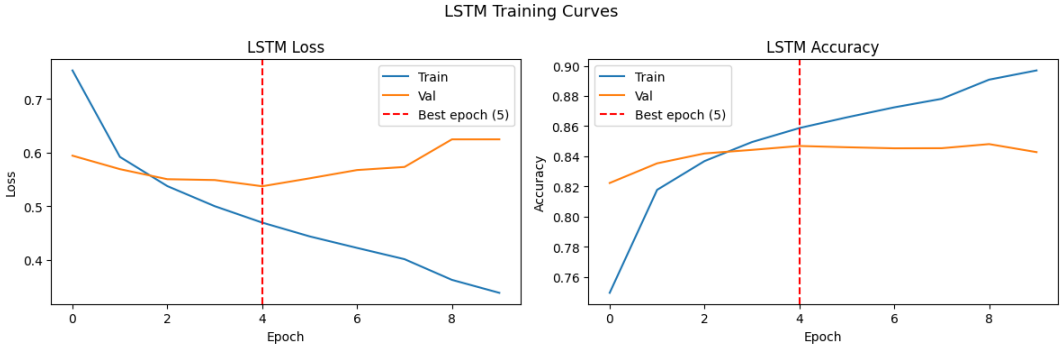

Fig. 3. LSTM training and validation loss and accuracy curves. Best checkpoint saved at epoch 5. Overfitting is evident from epoch 6 onward as training loss continues to decrease while validation loss rises.

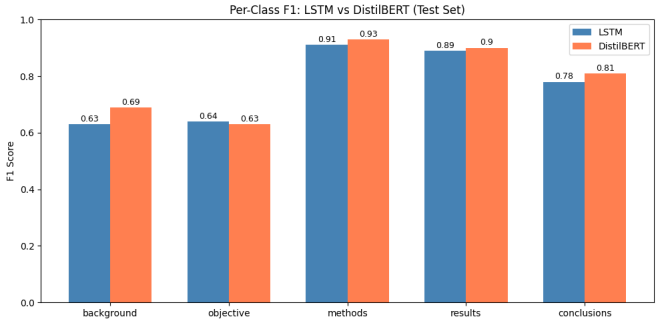

Fig. 4. Per-class F1 comparison between the LSTM baseline and DistilBERT on the test set. DistilBERT outperforms the LSTM on all five classes, with the largest gains on Background and Conclusions.

G. Clinical Relevance of Results

The per-class performance patterns observed in our results are clinically meaningful. Methods and Results sentences, which are most reliably classified by both models, correspond to the information most directly relevant to clinical decision-making: what was done and what was found. Background and Objective sentences, where both models perform worst, correspond to the framing and motivation of the study, which are less critical for rapid evidence synthesis. A clinical NLP pipeline built on either model could therefore be expected to reliably extract the most actionable portions of an RCT abstract, even if it occasionally mislabels introductory sentences. The persistent difficulty with Objective sentences highlights the importance of class weighting and minority class evaluation in any clinical NLP system trained on naturally imbalanced corpora [11].

V. Discussion & Limitations

A. Interpretation of Results

The results of this project demonstrate that both a lightweight bidirectional LSTM and a fine-tuned DistilBERT transformer can reliably classify sentences in RCT abstracts into their rhetorical roles, achieving 84% and 86% test accuracy respectively. The strong performance of both models on the Methods and Results classes confirms that these sentence types carry distinctive and learnable linguistic signatures, including procedural language, anonymized numeric tokens, and statistical terminology. The more modest performance on Background and Objective reflects a genuine semantic ambiguity between these classes that is not fully resolvable from sentence-level features alone, consistent with findings reported by Jin and Szolovits [7], who showed that modeling inter-sentence context within the abstract can further improve classification at these boundaries.

The most practically significant finding of this project is the efficiency tradeoff between the two models. DistilBERT achieves a 2 percentage point improvement in overall accuracy and a 0.02 improvement in macro F1 relative to the LSTM, at the cost of 10x more parameters and approximately 6x longer training time per epoch. In a research setting where compute resources are abundant and marginal performance gains are valuable, DistilBERT is the preferred choice. However, in a resource-constrained clinical deployment scenario, such as a hospital system running inference on a high-throughput stream of incoming abstracts, the LSTM offers a compelling alternative that is far cheaper to train, faster to run, and easier to interpret. This tradeoff mirrors findings in the broader clinical NLP literature, where lightweight models have repeatedly been shown to approach the performance of larger models on well-structured classification tasks [9].

The error analysis conducted in the demo notebook further supports these conclusions. Of the 29,578 test sentences, the two models disagreed on 3,109 predictions (10.5%). Qualitative inspection of disagreement cases revealed that the most common failure mode for both models is the Background-Objective boundary, where sentences introducing the study motivation or prior literature can plausibly belong to either class depending on subtle phrasing cues. Cases where neither model predicted correctly tended to involve genuine label ambiguity, such as concluding sentences that summarize results rather than making recommendations, or objective sentences phrased in a way that resembles background framing. These failure cases reflect real ambiguity in the annotation schema rather than

model deficiencies, and suggest that ceiling performance on this task may be lower than the 95%+ F1 scores reported by state-of-the-art hierarchical models [7] would imply for independent sentence classification.

B. Limitations

Several important limitations should be considered when interpreting these results. First, both models classify sentences independently, without access to the surrounding abstract context. Prior work has shown that modeling inter-sentence dependencies through hierarchical architectures can substantially improve performance at ambiguous class boundaries [7]. Our independent sentence classification approach was chosen for simplicity and practical deployability, but it sacrifices contextual information that could resolve many of the Background-Objective errors observed in our results.

Second, the PubMed RCT dataset is drawn exclusively from PubMed-indexed English-language RCT abstracts, which represent a specific and relatively homogeneous subset of biomedical literature. Models trained on this corpus may not generalize well to abstracts from other study designs such as observational studies or systematic reviews, to non-English literature, or to clinical documents with different structural conventions such as case reports or clinical notes. Any deployment of these models beyond the RCT abstract domain would require validation on representative out-of-domain data.

Third, both models were evaluated on a single train-validation-test split without cross-validation or bootstrapped confidence intervals. While the large size of the test set (29,578 sentences) provides reasonable statistical stability, we cannot formally quantify the uncertainty in our reported metrics. Future work should include confidence intervals computed via bootstrapping to support stronger comparative claims between models.

Fourth, the anonymization of numeric values with the `@` token, while consistent across the dataset, removes potentially informative features such as sample sizes, p-values, and effect sizes that could aid in distinguishing Results sentences from other classes. This is a dataset-level limitation that affects all models trained on this corpus equally.

C. Ethical Considerations

The deployment of NLP models in clinical and biomedical settings raises important ethical considerations that must be addressed before any real-world use. Bear Don't Walk et al. [11] identified training data bias, lack of model transparency, and risks of perpetuating health disparities as primary ethical concerns in clinical NLP. Although the PubMed RCT dataset is drawn from publicly available literature and does not contain individual patient data, it reflects the publication biases of the biomedical research community, including overrepresentation of studies conducted in high-income countries and underrepresentation of research on diseases and populations that are historically marginalized in clinical trial enrollment [12]. A model trained on this corpus may therefore perform less reliably on abstracts describing trials conducted in lower-resource settings or focused on underrepresented populations.

Model transparency is another important consideration. The LSTM model offers greater interpretability than DistilBERT, as its predictions can be more readily traced to specific vocabulary patterns learned during training. The DistilBERT model, while more accurate, operates as a relatively opaque function of its input tokens, making it harder for clinical users to understand or verify individual predictions. In any clinical deployment, model outputs should be presented as decision support tools rather than authoritative classifications, with final judgment retained by qualified clinicians or researchers [9].

Data privacy is not a direct concern for this project, as the PubMed RCT dataset contains only publicly available abstract text with no patient-identifiable information. However, any extension of this work to proprietary clinical data, such as unpublished trial reports or electronic health records, would require strict compliance with applicable data governance regulations including HIPAA in the United States.

Finally, the class imbalance present in the dataset raises fairness concerns at the model level. A system that systematically underperforms on Objective sentences, which represent study goals and research questions, could produce misleading structured summaries that omit or misrepresent the intended purpose of a clinical trial. This is particularly problematic in evidence synthesis applications where accurate representation of study objectives is critical to downstream clinical decision-making [15].

D. Future Work

Several directions could extend and improve this work. First, incorporating inter-sentence context through a hierarchical architecture, as proposed by Jin and Szolovits [7], would likely reduce Background-Objective confusion and improve overall macro F1. Second, fine-tuning a domain-specific pretrained model such as BioBERT [6] or PubMedBERT on this task could provide additional gains, particularly on boundary cases where general-domain language representations may lack the biomedical context needed to resolve ambiguity. Third, applying data augmentation techniques targeting the Objective minority class, such as synonym replacement or back-translation, could improve minority class recall without requiring additional labeled data. Finally, integrating this classification pipeline into a structured abstract generation system and evaluating it with clinical end users would provide valuable insight into its practical utility and usability in real-world evidence synthesis workflows.

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